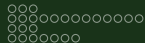


Monte Carlo Simulation

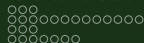
Kaiza Amouh

February 6, 2025



Outline

- 1 Monte Carlo Simulation
 - Monte Carlo Principle
 - Applications of Monte Carlo
- 2 Variance Reduction
 - Pseudo-Control variate
 - Antithetic method
 - Importance Sampling
- 3 Quasi Monte Carlo : low discrepancy sequences
 - Definition of Quasi-Random numbers
 - Most used low discrepancy sequences
- 4 American Monte Carlo



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- 1 Monte Carlo Simulation
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Principle of Monte Carlo Simulation

The Monte Carlo method is based on the Strong Law of Large Numbers (SLLN)

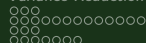
Strong Law of Large Numbers (SLLN)

If $(X_k)_{k \geq 1}$ denotes a sequence of independent copies of an integrable random variable X , defined on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$

$$\mathbb{P}(d\omega) - a.s. \quad \bar{X}_N(\omega) := \frac{X_1(\omega) + \dots + X_N(\omega)}{N} \xrightarrow{N \rightarrow +\infty} m_x := \mathbb{E}(X)$$

Important issues related to Monte Carlo

- What is the rate of convergence of the method ?
- How can the resulting error be controlled ?



Rate of convergence

- The (weak) rate of convergence in the *SLLN* is ruled by the *Central Limit Theorem (CLT)*

$$\sqrt{N} (\bar{X}_N - m_x) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma_x^2) \text{ as } N \rightarrow +\infty$$

$$\text{where } \sigma_x^2 = \text{Var}(X) := \mathbb{E} \left[(X - \mathbb{E}(X))^2 \right]$$

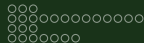
- The quadratic rate of convergence (i.e. the rate in $L^2(\mathbb{P})$) is exactly

$$\|\bar{X}_N - m_x\|_2 = \frac{\sigma_x}{\sqrt{N}}$$

a.s. rate of convergence : *Law of the Iterated Logarithm (LIL)*

Borel Cantelli

$$\limsup_{N \rightarrow +\infty} \sqrt{\frac{N}{2 \log(\log N)}} (\bar{X}_N - m_x) = \pm \sigma_x$$



Data-driven control of the error : confidence interval

- Using the *CLT*, for every real numbers $a < b$, we have

$$\lim_N \mathbb{P} \left[\sqrt{N} \left(\frac{\bar{X}_N - m_x}{\sigma_x} \right) \in [a, b] \right] \xrightarrow{N \rightarrow +\infty} \mathbb{P} (\mathcal{N}(0, 1) \in [a, b]) \\ = \Phi(b) - \Phi(a)$$

where Φ denotes the distribution function of the standard normal distribution

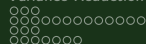
- Let $\alpha \in (0, 1)$ denote a *confidence level* (close to 1), and let a_α be the two-sided α -quantile defined as the unique solution to the equation

$$\mathbb{P} (|\mathcal{N}(0, 1)| \leq a_\alpha) = \alpha \quad \text{i.e.} \quad 2\Phi(a_\alpha) - 1 = \alpha$$

- Then one defines the random interval

**2 methods, compute directly
or fixe the confidence and simulate the N**

$$J_N^\alpha := \left[\bar{X}_N - a_\alpha \frac{\sigma_x}{\sqrt{N}}, \bar{X}_N + a_\alpha \frac{\sigma_x}{\sqrt{N}} \right]$$



Data-driven control of the error : confidence interval

95% alpha for instance

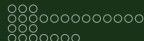
Probability of laying in the confidence interval

$$\mathbb{P}(m_x \in J_N^\alpha) = \mathbb{P}\left(\frac{\sqrt{N}}{\sigma_x} |\bar{X}_N - m_x| \leq a_\alpha\right) \xrightarrow{N \rightarrow +\infty} \mathbb{P}(|\mathcal{N}(0, 1)| \leq a_\alpha) = \alpha$$

- At this stage, σ_x is unknown and must be estimated
- Applying the *SLLN* to the sequence of integrable random variables $(X_k^2)_{k \geq 1}$ yields

$$\bar{V}_N := \frac{1}{N-1} \sum_{k=1}^N (X_k - \bar{X}_N)^2 \xrightarrow{N \rightarrow +\infty} \text{Var}(X) = \sigma_x^2$$

- We hence perform a *companion Monte Carlo simulation* to estimate the variance σ^2



Vanilla Option Pricing in a Black-Scholes model

The problem at Natixis: log normal vol for eqty and normal for the equity vol hence it was a problem

- A European *vanilla* option with maturity $T > 0$ is an option related to a European payoff $h_T := h(X_T)$ which only depends on X at time T
- In a complete market, the option premium at any time $t \in [0, T]$ is

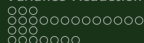
$$V_t = e^{-r(T-t)} \mathbb{E}(h(X_T) / \mathcal{F}_t)$$

Examples of Vanilla options with strike K

Vanila Call : $h(X_T) = (X_T - K)^+$

Best-of Call : $h(X_T, Y_T) = (\max(X_T, Y_T) - K)^+$

Exchange Call Spread : $h(X_T, Y_T) = ((X_T - Y_T) - K)^+$



Greeks computation in a Black-Scholes model

- The aim is to derive some representation of sensitivities as expectations
- These expectations can be evaluated using a Monte Carlo simulation in parallel with the premium computation
- Let $X_0 = x > 0$, and $X_t = x \exp \left[\left(r - \frac{\sigma^2}{2} \right) t + \sigma W_t \right]$
- Consider for every $x \in (0, \infty)$ the option price $f(x) := \mathbb{E}(h(X_T))$

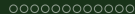
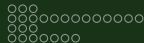
Proposition

- If $h : (0, +\infty) \rightarrow \mathbb{R}$ is differentiable and h' has a polynomial growth, then f is differentiable and

$$\forall x > 0, \quad f'(x) = \mathbb{E} \left(h'(X_T) \frac{X_T}{x} \right)$$

- If $h : (0, +\infty) \rightarrow \mathbb{R}$ is simply a Borel function with polynomial growth, then f is still differentiable and

$$\forall x > 0, \quad f'(x) = \mathbb{E} \left(h(X_T) \frac{W_T}{x \sigma T} \right)$$



Proof of the Proposition

- The first part can be obtained using the permutation of differentiation and integrals, together with the fact that

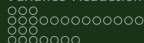
$$\frac{\partial}{\partial x} h(X_T) = h'(X_T) \frac{\partial X_T}{\partial x} = h'(X_T) \frac{X_T}{x}$$

- Under this assumption, let $\mu := r - \frac{\sigma}{2}$. We have:

$$\begin{aligned} f'(x) &= \int_{\mathbb{R}} h' \left(x e^{(\mu T + \sigma \sqrt{T} u)} \right) e^{(\mu T + \sigma \sqrt{T} u)} e^{-\frac{u^2}{2}} \frac{du}{\sqrt{2\pi}} \\ &= \frac{1}{x \sigma \sqrt{T}} \int_{\mathbb{R}} \frac{\partial h \left(x e^{(\mu T + \sigma \sqrt{T} u)} \right)}{\partial u} e^{-\frac{u^2}{2}} \frac{du}{\sqrt{2\pi}} \\ &= -\frac{1}{x \sigma \sqrt{T}} \int_{\mathbb{R}} h \left(x e^{(\mu T + \sigma \sqrt{T} u)} \right) \frac{\partial e^{-\frac{u^2}{2}}}{\partial u} \frac{du}{\sqrt{2\pi}} \end{aligned}$$

where we used an integration by part, taking advantage of the fact that : **due to the polynomial growth of h , we have:**

$$\lim_{|u| \rightarrow +\infty} h \left(x e^{(\mu T + \sigma \sqrt{T} u)} \right) e^{-\frac{u^2}{2}} = 0$$

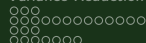


Proof of the Proposition

We can then write:

$$\begin{aligned}
 f'(x) &= \frac{1}{x\sigma\sqrt{T}} \int_{\mathbb{R}} h\left(xe^{(\mu T + \sigma\sqrt{T}u)}\right) ue^{-\frac{u^2}{2}} \frac{du}{\sqrt{2\pi}} \\
 &= \frac{1}{x\sigma T} \int_{\mathbb{R}} h\left(xe^{(\mu T + \sigma\sqrt{T}u)}\right) \sqrt{T}ue^{-\frac{u^2}{2}} \frac{du}{\sqrt{2\pi}} \\
 &= \frac{1}{x\sigma T} \mathbb{E}(h(X_T)W_T)
 \end{aligned}$$

this is to implement the delta for BS - and compare it to the Closed form delta



Computing the Value-at-Risk by Monte Carlo simulation

- Let X be a real-valued random variable, representative of a *loss*.
- For a given confidence level $\alpha \in (0, 1)$, the *Value-at-Risk* at level α is any real number $V@R_{\alpha, X}$ satisfying

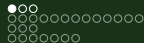
$$\mathbb{P}(X \leq V@R_{\alpha, X}) = \alpha \in (0, 1)$$

- For convenience, one often assumes that the lowest solution of the above equation is the $V@R_{\alpha, X}$
- One naive way to compute $V@R_{\alpha, X}$ is to estimate the empirical distribution function of a Monte Carlo simulation at some points ξ lying in a grid $\Gamma := \{\xi_i, i \in I\}$

$$\widehat{F}(\xi)_N := \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{\{X_k \leq \xi\}}, \quad \xi \in \Gamma$$

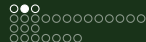
store your X_k so you can have it here

- Then one solves the equation $\widehat{F}(\xi)_N = \alpha$ (using an interpolation step of course)



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Variance Reduction - Motivations

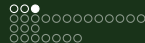
- For a large enough N , we have:

$$\mathbb{P}\left(m_x \in \left[\bar{X}_N - a_\alpha \frac{\sigma_x}{\sqrt{N}}, \bar{X}_N + a_\alpha \frac{\sigma_x}{\sqrt{N}}\right]\right) \approx 2\Phi(a_\alpha) - 1 = \alpha$$

- In numerical probability, we adopt a reverse point of view based on the needed accuracy $\epsilon > 0$
- To make $\bar{X}_N$ enter a confidence interval $[m_x - \epsilon, m_x + \epsilon]$ with a confidence level α , one needs to process a Monte Carlo simulation of size

$$N \geq N^x(\epsilon, \alpha) = \frac{a_\alpha^2 \text{Var}(X)}{\epsilon^2}$$

- For a given accuracy, $N^x(\epsilon, \alpha)$ grows linearly with the variance of X
- For a given variance, $N^x(\epsilon, \alpha)$ grows like the inverse of the square of the accuracy



Variance Reduction - (not so!) Naive approach

- Assume we know 2 random variables $X, X' \in L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ satisfying $\mathbb{E}(X) = \mathbb{E}(X') = m \in \mathbb{R}$
- If both X and X' can be simulated with an equivalent cost (complexity), then we choose the one with the lowest variance !

Practical implementation

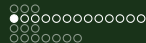
Assume there exists a random variable $Y \in L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ such that

- $\mathbb{E}(Y)$ can be computed at a very low cost by a deterministic method (closed form, numerical analysis method)
- The random variable $(X - Y)$ can be simulated with the same complexity as X
- The variance $\text{Var}(X - Y) < \text{Var}(X)$

Then the random variable $X' = X - Y + \mathbb{E}(Y)$ can be simulated instead of X , and we have

$$\mathbb{E}(X') = \mathbb{E}(X - Y) + \mathbb{E}(Y) = \overset{\text{target}}{\mathbb{E}(X)} \text{ and } \text{Var}(X') = \text{Var}(X - Y) < \text{Var}(X)$$

The random variable Y is called a *static control variate* for X



Variance Reduction - Pseudo Control Variate

- In option pricing, payoffs are **usually** non-negative
- For any random variable Y satisfying the two first conditions, together with $0 \leq Y \leq X$, then Y is a *good candidate* to reduce variance.
- The closer Y is to X , the smaller $(X - Y)$ is, and the smaller the variance is.

The *Jensen* inequality is an efficient tool to design pseudo-control variate, when dealing with path-dependent or multi-asset options.

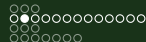
Jensen inequality

Let $X : (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R}$ be a random variable and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Suppose X and $g(X)$ are integrable. Then, for any sub- σ -field $\mathcal{B}$ of $\mathcal{A}$,

$$g(\mathbb{E}(X/\mathcal{B})) \leq \mathbb{E}(g(X)/\mathcal{B}) \quad \mathbb{P} - a.s.$$

In particular considering $\mathcal{B} = \{\emptyset, \Omega\}$ yields the above inequality for regular expectation i.e.

$$g(\mathbb{E}(X)) \leq \mathbb{E}(g(X))$$



Pseudo Control Variate - Example of Basket options

- We consider a call option on a basket of d risky assets, with strike K

$$h_T = \left(\sum_{i=1}^d \alpha_i X_T^i - K \right)^+$$

where $(X^1, \dots, X^d)$ models the price of d traded risky assets on a market and α_i are some positive weights satisfying $\sum_{1 \leq i \leq d} \alpha_i = 1$

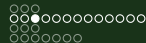
- Then the convexity of the exponential implies that

$$0 \leq e^{\sum_{1 \leq i \leq d} \alpha_i \log(X_T^i)} \leq \sum_{i=1}^d \alpha_i X_T^i$$

so that

$$h_T \geq h'_T := \left(e^{\sum_{1 \leq i \leq d} \alpha_i \log(X_T^i)} - K \right)^+ \geq 0$$

log normal



Pseudo Control Variate - Example of Basket options

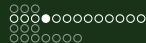
- The motivation for this example is that a (possibly correlated) d -dimensional Black-Scholes model, $\sum_{1 \leq i \leq d} \alpha_i \log(X_T^i)$ still has a normal distribution.
- Therefore, the Call-like European option h'_T has a closed form.
- The correlated d -dimensional Black-Scholes model can be defined by the following system of SDE's:

$$dX_t^i = X_t^i \left(r dt + \sum_{j=1}^i \sigma_{ij} dW_t^j \right), \quad t \in [0, T], \quad X_0^i := x_i > 0, \quad i = 1, \dots, d$$

where $W = (W^1, \dots, W^d)$ is a standard d -dimensional Brownian motion and $\sigma = [\sigma_{ij}]_{1 \leq i, j \leq d}$ is a lower triangular matrix.

- Its closed form solution is given by:

$$X_t^i = x_i \exp \left(\left(r - \frac{\sigma_{i,i}^2}{2} \right) t + \sum_{j=1}^i \sigma_{ij} W_t^j \right), \quad \text{where} \quad \sigma_{i,\cdot}^2 = \sum_{j=1}^i \sigma_{ij}^2$$



Example of Basket options - Implementation

- Step 1 : Compute analytically the expectation $\mathbb{E}(e^{-rT} h'_T)$

$$\sum_{1 \leq i \leq d} \alpha_i \log(X_T^i / x_i) \stackrel{d}{=} \mathcal{N} \left(\left(r - \frac{1}{2} \sum_{1 \leq i \leq d} \alpha_i \sigma_i^2 \right) T ; \alpha' \sigma \sigma' \alpha T \right)$$

where α is the column vector with components $\alpha_i, i = 1, \dots, d$

$$e^{-rT} \mathbb{E}(h'_T) = \text{Call}_{BS} \left(\prod_{i=1}^d x_i^{\alpha_i}, K, \left(r - \frac{1}{2} \sum_{1 \leq i \leq d} \alpha_i \sigma_i^2 + \frac{1}{2} \alpha' \sigma \sigma' \alpha \right), \sqrt{\alpha' \sigma \sigma' \alpha}, T \right)$$

- Step 2 : Joint simulation of the couple (h_T, h'_T)

$$e^{-rT} (h_T - h'_T) = e^{-rT} \left(\left(\sum_{i=1}^d \alpha_i X_T^i - K \right)^+ - \left(e^{\sum_{1 \leq i \leq d} \alpha_i \log(X_T^i)} - K \right)^+ \right)$$

Pseudo Control Variate - Example of Asian options

$$\text{Let } X_t = X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma W_t \right) \text{ and } h_T = \phi \left(\frac{1}{T} \int_0^T X_t dt \right)$$

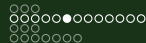
h_T is a generic Asian payoff with ϕ is non-negative, and X_t a regular Black-Scholes dynamics

The (standard) Jensen inequality applied to the probability measure $\frac{1}{T} \mathbf{1}_{[0,T]}(t)dt$ implies

$$\begin{aligned} \frac{1}{T} \int_0^T X_t dt &\geq X_0 \exp \left(\frac{1}{T} \int_0^T \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma W_t \right) dt \right) \\ &= X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) \frac{T}{2} + \frac{\sigma}{T} \int_0^T W_t dt \right) \end{aligned}$$

Now

$$\int_0^T W_t dt = TW_T - \int_0^T s dW_s = \int_0^T (T-s) dW_s$$



Pseudo Control Variate - Example of Asian options

Then we have

$$\frac{1}{T} \int_0^T W_t dt \stackrel{d}{=} \mathcal{N} \left(0, \frac{1}{T^2} \int_0^T s^2 ds \right) \stackrel{d}{=} \mathcal{N} \left(0, \frac{T}{3} \right)$$

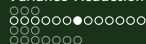
This suggests to rewrite the right hand side of the above inequality in a "Black-Scholes asset" style, namely:

$$\frac{1}{T} \int_0^T X_t dt \geq X_0 e^{-\left(\frac{r}{2} + \frac{\sigma^2}{12}\right)T} \exp \left(\left(r - \frac{\sigma^2}{6} \right) T + \sigma \frac{1}{T} \int_0^T W_t dt \right)$$

This naturally leads to introduce the so-called *Kemna-Vorst* pseudo-control variate

$$h_T^{KV} := \phi \left(X_0 e^{-\left(\frac{r}{2} + \frac{\sigma^2}{12}\right)T} \exp \left(\left(r - \frac{\sigma^2}{6} \right) T + \sigma \frac{1}{T} \int_0^T W_t dt \right) \right)$$

which is clearly of Black-Scholes type and satisfies $h_T \geq h_T^{KV}$



Example of Asian options - Implementation

- Step 1 : The random variable h_T^{KV} is an admissible control variate as soon as the vanilla option related to the payoff $\phi(X_T)$ has a closed form

$$e^{-rT} \mathbb{E}(\phi(X_T)) = \text{premium}_{BS}^{\phi}(X_0, T, r, \sigma)$$

then we have

$$e^{-rT} \mathbb{E}(h_T^{KV}) = \text{premium}_{BS}^{\phi} \left(X_0 e^{-\left(\frac{r}{2} + \frac{\sigma^2}{12}\right)T}, T, r, \frac{\sigma}{\sqrt{3}} \right)$$

- Step 2 : One has to simulate independent copies of the couple (h_T, h_T^{KV}) , i.e. the couple

$$\left((W_t)_{t \in [0, T]}, \quad \frac{1}{T} \int_0^T W_t dt \right)$$

Example of Asian options - Computation of time integrals

- In order to perform this simulation, one needs to compute time integrals in both payoffs
- Numerical integration methods are applied, such as the mid-point quadrature formula

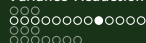
$$\frac{1}{T} \int_0^T f(t) dt \approx \frac{1}{n} \sum_{k=1}^n f\left(\frac{2k-1}{2n} T\right)$$

- Keep in mind that the functions f of interest are here given by

$$f(t) = \phi \left(X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma W_t(\omega) \right) \right) \quad \text{or} \quad f(t) = W_t(\omega)$$

for the first and the second payoff function respectively.

- Their regularity is α -Holder, $\alpha < \frac{1}{2}$ (locally for the payoff h_T)



Pseudo Control Variate - Example of Best-of-call options

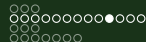
- The payoff of a *Best-of-call* option is given by

$$h_T = (\max(X_T^1, X_T^2) - K)^+$$

- Using the convexity inequality (an application of Jensen inequality)

$$\sqrt{ab} \leq \max(a, b) \implies h_T \geq h'_T := \left(\sqrt{X_T^1 X_T^2} - K \right)^+$$

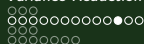
- In a 2-dimensional Black-Scholes model, the option with payoff h'_T has a closed form
- The procedure can be improved by noting that more generally, $a^\theta b^{1-\theta} \leq \max(a, b)$ when $\theta \in (0, 1)$



Using parity equations to produce control variates

- In derivatives pricing, one can always build a control variate as soon as the payoff of interest satisfies a so-called *parity equation*
- These *parity equations* are model free, so they can be applied for various specifications of the dynamics of the underlying asset.
- We denote by $(S_t)_{t \in [0, T]}$ the risky asset
- We work under the risk-neutral probability:

$(e^{-rt} S_t)_{t \in [0, T]}$ is a martingale on the scenarii space $(\Omega, \mathcal{A}, \mathbb{P})$



Vanilla Call-Put parity ($d = 1$)

- Let the premia of *Call* and *Put* option be denoted as follows

$$Call_0 = e^{-rT} \mathbb{E} \left((S_T - K)^+ \right) \quad \text{and} \quad Put_0 = e^{-rT} \mathbb{E} \left((K - S_T)^+ \right)$$

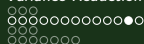
- Since $(S_T - K)^+ - (K - S_T)^+ = S_T - K$ and $(e^{-rt} S_t)_{t \in [0, T]}$ is a martingale,

$$Call_0 - Put_0 = S_0 - e^{-rT} K$$

- So that $Call_0 = \mathbb{E}(X) = \mathbb{E}(X')$ with

$$X := e^{-rT} (S_T - K)^+ \quad \text{and} \quad X' := e^{-rT} (K - S_T)^+ + S_0 - e^{-rT} K$$

- As a result, one sets $Y = X - X' = e^{-rT} S_T - S_0$
- Y turns out to be the terminal value of a martingale null at time 0 !



Asian Call-Put parity ($d = 1$)

- We assume that the averaging phase runs from T_0 to T

$$Call_0^{As} = e^{-rT} \mathbb{E} \left(\left(\frac{1}{T - T_0} \int_{T_0}^T S_t dt - K \right)^+ \right)$$

$$Put_0^{As} = e^{-rT} \mathbb{E} \left(\left(K - \frac{1}{T - T_0} \int_{T_0}^T S_t dt \right)^+ \right)$$

- Using *Fubini's Theorem* and the fact that $(e^{-rt} S_t)_{t \in [0, T]}$ is a martingale, we have:

$$Call_0^{As} - Put_0^{As} = S_0 \frac{1 - e^{-r(T - T_0)}}{r(T - T_0)} - e^{-rT} K$$

Asian Call-Put parity ($d = 1$)

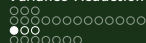
- Once again we can write $Call_0^{As} = \mathbb{E}(X) = \mathbb{E}(X')$ with

$$X := e^{-rT} \left(\frac{1}{T - T_0} \int_{T_0}^T S_t dt - K \right)^+$$

$$X' := S_0 \frac{1 - e^{-r(T-T_0)}}{r(T - T_0)} - e^{-rT} K + e^{-rT} \left(K - \frac{1}{T - T_0} \int_{T_0}^T S_t dt \right)^+$$

- Which leads to

$$Y = e^{-rT} \frac{1}{T - T_0} \int_{T_0}^T S_t dt - S_0 \frac{1 - e^{-r(T-T_0)}}{r(T - T_0)}$$



Antithetic method : negatively correlated variables

- We now assume that X and Y have not only the same expectation m_X but also the same variance
- If X and Y are negatively correlated, we set

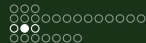
$$\chi := \frac{X + Y}{2}$$

- We have:

$$\text{Var}(\chi) = \frac{1}{4} (\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)) = \frac{\text{Var}(X) + \text{Cov}(X, Y)}{2}$$

- The size $N^X(\epsilon, \alpha)$ and $N^\chi(\epsilon, \alpha)$ of the simulation using X and χ respectively, to enter a given interval $[m - \epsilon, m + \epsilon]$ is given by

$$N^X = \left(\frac{a_\alpha}{\epsilon}\right)^2 \text{Var}(X) \quad \text{and} \quad N^\chi = \left(\frac{a_\alpha}{\epsilon}\right)^2 \text{Var}(\chi)$$



Antithetic method : complexity analysis

- Let κ be the complexity for simulating X . Simulating χ hence requires 2κ complexity **computing 2**
- One would prefer simulating χ only if $2\kappa N^X < \kappa N^X \Leftrightarrow \text{Cov}(X, Y) < 0$

Proposition : Antithetic generation

- Let $\phi, \psi : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be two monotone functions with the same monotony

Let $T : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a nonincreasing transform and let $Z : (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow \mathbb{R}$ be a random variable.

Assume that $\phi(Z), \psi(Z), \psi(T(Z)) \in L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$. Then

$$\text{Cov}(\phi(Z), \psi(Z)) \geq 0 \quad \text{and} \quad \text{Cov}(\phi(Z), \psi(T(Z))) \leq 0$$

- If furthermore $\psi = \phi$ and $Z \stackrel{d}{=} T(Z)$, then $X = \phi(Z)$ and $Y = \phi(T(Z))$ are identically distributed and satisfy $\text{Cov}(X, Y) \leq 0$

In that case, the random variables X and Y are called **antithetic**

Antithetic method : Application

you simulate 5k instead of 10k and you look at W and $-W$ you get

The antithetic ^{10k} random variable method applies in two situations

- The symmetric random variable $Z : \longrightarrow Z \stackrel{d}{=} -Z$ i.e. $T(z) = -z$
- The $[0, L]$ -valued random variable Z such that $Z \stackrel{d}{=} L - Z$ i.e. $T(z) = L - z$

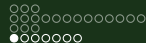
European option pricing in a B&S model

For $Z \rightsquigarrow \mathcal{N}(0, 1)$, the payoff is $h_T = \phi(X_T) = \phi\left(X_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z}\right)$

The function $z \mapsto \phi\left(X_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z}\right)$ is monotone, and $W_T \stackrel{d}{=} -W_T$

Uniform distribution on the unit interval : $U \rightsquigarrow \mathcal{U}([0, 1])$

If ϕ is monotone on $[0, 1]$, then $\text{Var}\left(\frac{\phi(U) + \phi(1-U)}{2}\right) \leq \frac{1}{2} \text{Var}(\phi(U))$



Basic principle of importance sampling

- Let $X : (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow (E, \mathcal{E})$ be an E -valued random variable?
- Let μ be a σ -finite *reference* measure on $(E, \mathcal{E})$ so that

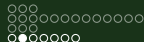
$\exists$ a density $f : (E, \mathcal{E}) \rightarrow (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$ such that $\mathbb{P}_X = f \cdot \mu$

- In practice, we will assume $E = \mathbb{R}$ and μ =Lebesgue measure
- Let $h \in L^1(\mathbb{P}_X)$. Then,

$$\mathbb{E}(h(X)) = \int_{\mathbb{R}} h(x) \mathbb{P}_X(dx) = \int_{\mathbb{R}} h(x) f(x) \mu(dx)$$

- For any μ -a.s. positive probability density g defined on $(\mathbb{E}, \mu)$, one has

$$\mathbb{E}(h(X)) = \int_{\mathbb{R}} h(x) f(x) \mu(dx) = \int_{\mathbb{R}} \frac{h(x) f(x)}{g(x)} g(x) \mu(dx) = \mathbb{E} \left(\frac{h(Y) f(Y)}{g(Y)} \right)$$

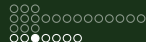


Importance sampling's variance reduction

$$\begin{aligned}
 \mathbb{E} \left[\left(\frac{h(Y)f(Y)}{g(Y)} \right)^2 \right] &= \mathbb{E} \left[\left(\frac{hf}{g}(Y) \right)^2 \right] \\
 &= \int_{\mathbb{R}} \left(\frac{h(x)f(x)}{g(x)} \right)^2 g(x) \mu(dx) \\
 &= \int_{\mathbb{R}} h(x)^2 \frac{f(x)}{g(x)} f(x) \mu(dx) \\
 &= \mathbb{E} \left(h(X)^2 \frac{f}{g}(X) \right)
 \end{aligned}$$

- Simulating $\frac{hf}{g}(Y)$ rather than $h(X)$ will reduce the variance if and only if

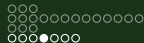
$$\mathbb{E} \left(h(X)^2 \frac{f}{g}(X) \right) < \mathbb{E} \left(h(X)^2 \right)$$



How to design and implement importance sampling ?

- The goal is to replace X by Y so that $\frac{h}{fg}(Y)$ is "closer" than $h(X)$ to their common mean
- Consider a *Call* with strike K and $X_0 \ll K$ (deep out-of-the-money)
- Most scenarios ω would yield $(X_T(\omega) - K)^+ = 0$
- Small number of events with positive payoffs $\Rightarrow$ rough estimation of $\mathbb{E}((X_T - K)^+)$
- If we can switch from $(X_t)_{t \in [0, T]}$ to $(Y_t)_{t \in [0, T]}$ so that Y_T takes most of its values in $[K, +\infty)$, then

$$\mathbb{E}((X_T - K)^+) = \mathbb{E}\left((Y_T - K)^+ \frac{f}{g}(Y_T)\right)$$

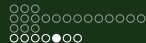


Parametric importance sampling

- We introduce a family of random variables $(Y_\theta)_{\theta \in \Theta}$ such that $g_\theta > 0$
- Assume $\exists \theta_0 \in \Theta$ such that $Y_{\theta_0} = X$
- The problem becomes a parametric optimization problem

$$\min_{\theta \in \Theta} \left\{ \mathbb{E} \left[\left(h(Y_\theta) \frac{f}{g_\theta}(Y_\theta) \right)^2 \right] = \mathbb{E} \left(h(X)^2 \frac{f}{g_\theta}(X) \right) \right\}$$

- Of course, there is no reason why the solution to the above problem should be θ_0 (unless the parametric model is inappropriate)



Example with Cameron-Martin formula

- In the 1-dimensional Black-Scholes model, the premium of an option with payoff ϕ is

$$e^{-rT} \mathbb{E}(\phi(X_T)) = \mathbb{E}(h(Z)) = \int_{\mathbb{R}} h(z) e^{-\frac{z^2}{2}} \frac{dz}{\sqrt{2\pi}}$$

where $Z \stackrel{d}{=} \mathcal{N}(0, 1)$ and $h(z) = e^{-rT} \phi \left(X_0 e^{(r - \frac{\sigma^2}{2})T + \sigma \sqrt{T}z} \right)$

- From now on, we will focus on

$$\mathbb{E}(h(Z)) = \int_{\mathbb{R}} h(z) f(z) dz \quad \text{with a minus} \quad \text{where} \quad f(z) = \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$$

- The idea is to introduce the parametric family

$$Y_{\theta} = Z + \theta, \quad \theta \in \Theta := \mathbb{R}$$

Example with Cameron-Martin formula

With $g_\theta(y) = \frac{e^{-\frac{(y-\theta)^2}{2}}}{\sqrt{2\pi}}$, we have $\frac{f}{g_\theta}(y) = e^{-\theta y + \frac{\theta^2}{2}}$

from Keyman HJB then solve PDE with optimal come back, this is baby method of that

Cameron-Martin formula

$$\begin{aligned}\mathbb{E}(h(Z)) &= e^{\frac{\theta^2}{2}} \mathbb{E}(h(Y_\theta)e^{-\theta Y_\theta}) = e^{\frac{\theta^2}{2}} \mathbb{E}(h(Z + \theta)e^{-\theta(Z+\theta)}) \\ &= e^{-\frac{\theta^2}{2}} \mathbb{E}(h(Z + \theta)e^{-\theta Z})\end{aligned}$$

The next step is to choose a "good" θ which significantly reduces the variance

$$\min_{\theta \in \mathbb{R}} \left[e^{\frac{\theta^2}{2}} \mathbb{E}(h^2(Z)e^{-\theta Z}) = e^{-\theta^2} \mathbb{E}(h^2(Z + \theta)e^{-2\theta Z}) \right]$$

Heuristic Sub-optimal approaches

We could "re-center" the simulation of X around K by replacing Z by $Z + \theta$

expectation of the underlying dancing around K instead of X_0

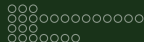
$$\mathbb{E} \left(X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} (Z + \theta) \right) \right) = K \implies \theta := - \frac{\log(X_0/K) + rT}{\sigma \sqrt{T}}$$

Or solve the similar, although slightly different equation

$$\mathbb{E} \left(X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} (Z + \theta) \right) \right) = e^{-rT} K \implies \theta := - \frac{\log(X_0/K)}{\sigma \sqrt{T}}$$

Or search for θ such that $\mathbb{P} \left(X_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} (Z + \theta) \right) < K \right) = \frac{1}{2}$

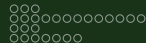
$$\implies \theta := - \frac{\log(X_0/K) + \left(r - \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}}$$



Outline

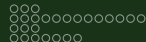
Sometimes one method is enough and adding other doesn't help (he wants to see that in the project application but not always the case)

- 1 Monte Carlo Simulation
- 2 Variance Reduction
- 3 Quasi Monte Carlo : low discrepancy sequences
 - Definition of Quasi-Random numbers
 - Most used low discrepancy sequences
- 4 American Monte Carlo



Quasi-Random Numbers

- Quasi-Random numbers are generated through purely deterministic sequences, and these sequences don't even attempt to emulate the behavior of *independent uniform random variables*.
- They rather cover the space in d dimensions with fewer gaps than independent random variables would normally.
- Pseudo-Random Numbers may result in large variance when used in a Monte Carlo simulation
- The so-called Quasi-Random Numbers present excellent properties in terms of variance reduction.
- The essential property of such a sequence is the so-called Low Discrepancy



Discrepancy of a Sequence

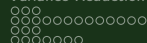
Let $(\xi_n)_{n \geq 1}$ be a $[0, 1]^d$ -valued sequence. One defines the discrepancy of (ξ_n) as follows:

Discrepancy at the origin or "Star Discrepancy"

$$D_n^*(\xi) := \sup_{x \in [0, 1]^d} \left| \frac{1}{n} \sum_{k=1}^n \mathbf{1}_{[0, x]}(\xi_k) - \overset{\text{lebesgue measure}}{\prod_{i=1}^d x^i} \right|$$

Extreme Discrepancy

$$D_n^\infty(\xi) := \sup_{x, y \in [0, 1]^d} \left| \frac{1}{n} \sum_{k=1}^n \mathbf{1}_{[x, y]}(\xi_k) - \prod_{i=1}^d (y^i - x^i) \right|$$



Portemanteau Theorem

The following assertions are equivalent

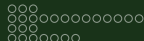
(i) $(\xi_{x \geq 1})$ is uniformly distributed on $[0, 1]^d$

(ii) For every $x \in \llbracket 0, 1 \rrbracket := [0, 1]^d$,

$$\frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\llbracket 0, x \rrbracket} (\xi_k) \rightarrow \lambda_d (\llbracket 0, x \rrbracket) := \prod_{i=1}^d x^i \quad \text{as } n \rightarrow +\infty$$

(iii) $D_n^\infty (\xi) \rightarrow 0 \quad \text{as } n \rightarrow +\infty$

(iv) $D_n^* (\xi) \rightarrow 0 \quad \text{as } n \rightarrow +\infty$



Portemanteau Theorem

Theoreme chinois

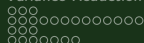
The following assertions are equivalent

- (i) $(\xi_{x \geq 1})$ is uniformly distributed on $[0, 1]^d$
- (ii) (Weyl's criterion) For every integer $p \in \mathbb{N}^d \setminus \{0\}$

$$\frac{1}{n} \sum_{k=1}^n e^{2i\pi(p|\xi_k)} \rightarrow 0 \quad \text{as } n \rightarrow +\infty \quad (\text{where } i^2 = -1).$$

- (iii) (Bounded Riemann integrable function) For every bounded λ_d -a.s. continuous Lebesgue-measurable function $f : [0, 1]^d \rightarrow \mathbb{R}$

$$\frac{1}{n} \sum_{k=1}^n f(\xi_k) \rightarrow \int_{[0,1]^d} f(x) \lambda_d(dx) \quad \text{as } n \rightarrow +\infty$$



Van der Corput and Halton sequence

Let $p_1, \dots, p_d$ be the first d prime numbers (or simply, d pairwise prime numbers).

the d -dimensional Halton sequence is defined, for every $n \geq 1$, by:

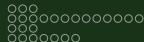
$$\xi_n = (\Phi_{p_1}(n), \dots, \Phi_{p_d}(n))$$

where the so-called "*radical inverse functions*" Φ_p are defined by

$$\Phi_p(n) = \sum_{k=0}^r \frac{a_k}{p^{k+1}} \quad \text{with } n = a_0 + a_1 p + \dots + a_r p^r$$

r is the last operation of division

and $a_i \in \{0, \dots, (p-1)\}$, $a_r \neq 0$, denotes the p -adic expansion of n



Van der Corput and Halton sequence

The discrepancy of Halton sequence can be bounded using the *Chinese Remainder Theorem* (a.k.a *Theoreme chinois* in French)

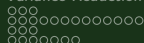
$$D_n^*(\xi) \leq \frac{1}{n} \left(d + \prod_{i=1}^d \left((p_i - 1) \frac{\log n}{2 \log p_i} + \frac{p_i + 2}{2} \right) \right), \quad n \geq 1$$

Van der Corput sequence ($d = 1$)

When $d = 1$, the sequence $(\Phi_p(n))_{n \geq 1}$ is called the *the p -adic Van der Corput sequence* (and the integer p needs not to be prime).

For $p = 2$, one easily checks that the first terms of the VdC(2) sequence are as follows

$$\xi_1 = \frac{1}{2}, \xi_2 = \frac{1}{4}, \xi_3 = \frac{3}{4}, \xi_4 = \frac{1}{8}, \xi_5 = \frac{5}{8}, \xi_6 = \frac{3}{8}, \xi_7 = \frac{7}{8} \dots$$



The Kakutani sequences

Based on Kakutani adding machine : p -adic addition on $[0, 1]$
the fastest and handle well dilensions but doesn't actually make random simulations

Principle of p -adic addition on $[0, 1]$

A p -adic addition is a binary operation defined on the set of p -adic expansions.

Let $\oplus_p$ denote this addition

If $x, y \in [0, 1]$ have their respective regular p -adic expansions as

$$x = \overline{0, x_1 x_2 \dots x_k \dots}^p \quad \text{and} \quad y = \overline{0, y_1 y_2 \dots y_k \dots}^p$$

Then

$$(x \oplus_p y)_k = (x_k + y_k) \mathbf{1}_{\{x_{k-1} + y_{k-1} \leq p-1\}} + (1 + x_k + y_k) \mathbf{1}_{\{x_{k-1} + y_{k-1} \geq p\}}$$

With the convention $x_1 = y_1 = 0$

Convention: $1 = \overline{0.(p-1)(p-1)(p-1)\dots}^p$

Example: $0.12333\dots \oplus_{10} 0.412777\dots = 0.535011\dots$

The Kakutani sequences

For every $y \in [0, 1]$, one defines the associated p -adic rotation with angle y by $T_{p,y}(x) := x \oplus_p y$ ***y is your theta, and the theta is from a p-adic number***

Construction of the sequence

Let $p_1, \dots, p_d$ denote the first d prime numbers, $y_1, \dots, y_d \in (0, 1)$, where y_i is a p_i -adic rational number satisfying $y_i \geq \frac{1}{p_i}$, $i = 1, \dots, d$ and $x_1, \dots, x_d \in [0, 1]$.

Then the Kakutani sequence $(\xi)_{n \geq 1}$ is defined by

$$\xi_n := (T_{p_i, y_i}^{n-1}(x_i))_{1 \leq i \leq d}, \quad n \geq 1$$

$$D_n^*(\xi) \leq \frac{1}{n} \left(1 + \prod_{i=1}^d \left((p_i - 1) \left\lfloor \frac{\log(p_i n)}{\log(p_i)} \right\rfloor \right) \right) = O\left(\frac{(\log n)^d}{n}\right) \text{ as } n \rightarrow +\infty$$

The Faure sequences

Let p be the smallest prime integer not lower than d (i.e. $p \geq d$).

d -dimensional Faure sequence for every $n \geq 1$

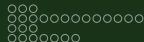
$$\xi_n = (\Phi_p(n-1), C_p(\Phi_p(n-1), \dots, C_p^{d-1}(\Phi_p(n-1))))$$

where Φ_p still denotes the p -adic radical inverse function and, for every p -adic rational number u with (regular) p -adic expansion $u = \sum_{k \geq 0} u_k p^{-(k+1)} \in [0, 1]$

$$C_p(u) = \sum_{k \geq 0} \underbrace{\left(\sum_{j \geq k} \binom{j}{k} u_j \text{ mod. } p \right)}_{\in \{0, \dots, (p-1)\}} p^{-(k+1)}$$

The discrepancy at the origin satisfies

$$D_n^*(\xi) \leq \frac{1}{n} \left(\frac{1}{d!} \left(\frac{p-1}{2 \log p} \right)^d (\log n)^d + O((\log n)^{d-1}) \right)$$



The Niederreiter sequences

Let $q \geq d$ be the smallest primary integer not lower than d (a primary integer reads $q = p^r$ with p prime).

The $(0, d)$ -Niederreiter sequence is defined for $n \geq 1$ by:

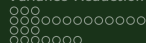
$$\xi_n = (\Psi_{q,1}(n-1), \Psi_{q,2}(n-1), \dots, \Psi_{q,d}(n-1))$$

where

$$\Psi_{q,i}(n) := \sum_j \psi^{j-1} \left(\sum_k C_{(j,k)}^{(i)} \Psi(a_k) \right) q^{-j}$$

and $\Psi : \{0, \dots, (q-1)\} \rightarrow \mathbb{F}_q$ is a one-to-one correspondence between $\{0, \dots, (q-1)\}$ and the finite field $\mathbb{F}_q$ with cardinal q satisfying $\Psi(0) = 0$ and

$$C_{(j,k)}^{(i)} = \binom{k}{j-1} \Psi(i-1)$$



Particular cases of Niederreiter sequences

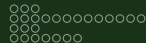
Sobol sequence

When $q = 2^r$, with $2^{r-1} < d \leq 2^r$, the Neiderreiter sequence coincides with Sobol's sequence

Faure sequence

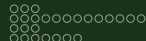
When p is the lowest prime number not lower than d , the Neiderreiter sequence coincides with the Faure sequence

- The sequences of this family all have discrepancy satisfying an upper bound with a structure similar to that of the Faure sequence.



Outline

- ① Monte Carlo Simulation
- ② Variance Reduction
- ③ Quasi Monte Carlo : low discrepancy sequences
- ④ American Monte Carlo



Bermudan options

- Let us consider $0 = t_0 < t_1 < \dots < t_n = T$ a discrete subdivision of $[0, T]$
- A Bermudan option gives the right to the buyer to exercise at any date $t_0, \dots, t_n$ and pays $f(S_{t_k})$ at time t_k
- Let $(\tilde{V}_t)_{0 \leq t \leq T}$ denote the associated hedging portfolio. Then :
 - at date $T = t_n$ we have $V_{t_n} = f(S_{t_n})$
 - at date $T = t_{n-1}$ we have

$$\begin{aligned} V_{t_{n-1}} &= \max \left[f(S_{t_{n-1}}) ; e^{-r(t_n - t_{n-1})} \mathbb{E} \left(f(S_{t_n}) / \mathcal{F}_{t_{n-1}} \right) \right] \\ &= \max \left[f(S_{t_{n-1}}) ; e^{-r(t_n - t_{n-1})} \mathbb{E} \left(V_{t_n} / \mathcal{F}_{t_{n-1}} \right) \right] \end{aligned}$$

- The same way we get $\forall k \in \{0, \dots, n-1\}$

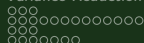
$$\begin{cases} V_T &= f(S_T) \\ V_{t_k} &= \max \left[f(S_{t_k}) ; e^{-r(t_{k+1} - t_k)} \mathbb{E} \left(V_{t_{k+1}} / \mathcal{F}_{t_k} \right) \right] \end{cases}$$

Remarks

- Note that the process $(S_t)_{0 \leq t \leq T}$ is Markovian then

$$\mathbb{E}(V_{t_{k+1}}/\mathcal{F}_{t_k}) = \mathbb{E}(V_{t_{k+1}}/S_{t_k})$$

- A Bermudan option is more expensive than an European option.
- If we let $n \rightarrow +\infty$, then the price of the Bermudan option tends to the price of an American option



Martingale Stopping theorem

Now if we consider $\forall k \in \{0, \dots, n-1\}$

$$\begin{cases} V_T &= f(S_T) \\ V_{t_k} &= \max \left[f(S_{t_k}) ; e^{-r(t_{k+1}-t_k)} \mathbb{E} (V_{t_{k+1}} / \mathcal{F}_{t_k}) \right] \end{cases}$$

Then

$$V_{t_k} = \sup_{\tau \in \{t_k, \dots, t_n\}} e^{-r(\tau-t_k)} \mathbb{E} [f(S_\tau) / S_{t_k}]$$

Theorem

The stopping time

$$\tau_k^* = \inf \left\{ t_i \in \{t_k, \dots, t_n\} \mid f(S_{t_i}) \geq e^{-r(t_{i+1}-t_i)} \mathbb{E} [f(S_{t_{i+1}}) / S_{t_k}] \right\}$$

satisfies

$$V_{t_k} = e^{-r(\tau_k^*-t_k)} \mathbb{E} [f(S_{\tau_k^*}) / S_{t_k}]$$

Longstaff-Schwarz algorithm

- The sequence $(\tau_k^*)_{0 \leq k \leq n}$ satisfies the dynamic programming principle

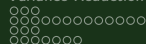
$$\begin{cases} \tau_n^* &= T \\ \tau_k^* &= t_k \mathbf{1}_{B_k} + \tau_{k+1}^* \mathbf{1}_{B_k^c}, \text{ for } 0 \leq k \leq n-1 \end{cases}$$

where

$$B_k = \left\{ f(S_{t_k}) \geq \mathbb{E} \left[e^{-r(\tau_{k+1}^* - t_k)} f(S_{\tau_{k+1}^*}) / S_{t_k} \right] \right\}$$

- The expectation can be approximated by a regression, based on a basis function $(P_l)_{l \geq 1}$

$$\mathbb{E} \left[e^{-r(\tau_{k+1}^* - t_k)} f(S_{\tau_{k+1}^*}) / S_{t_k} \right] = \sum_{l \geq 1} \alpha_{k,l} P_l(S_{t_k})$$



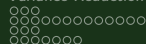
Regression

- Using the definition of the conditional expectation, the sequence $(\alpha_{k,l})_{l \geq 1}$ is the sequence that minimizes the distance

$$\mathbb{E} \left[\left(e^{-r(\tau_{k+1}^* - t_k)} f(S_{\tau_{k+1}^*}) - \sum_{l \geq 1} \alpha_{k,l} P_l(S_{t_k}) \right)^2 \right]$$

- In practice, we need to truncate the sum $\sum_{l \geq 1} \alpha_{k,l} P_l(S_{t_k})$ and approximate it by

$$\sum_{l \geq 1}^L \alpha_{k,l} P_l(S_{t_k}), \text{ where } L > 1$$



Longstaff-Schwartz algorithm

- Simulate $(S_{t_0}^j, \dots, S_{t_n}^j)_{1 \leq j \leq M}$, M copies of $(S_{t_0}, \dots, S_{t_n})$
- For all $1 \leq j \leq M$ we set $\tau_{j,n} := t_n = T$
- Then compute the sequence $(\alpha_{k,l}^j)_{1 \leq l \leq L}$ that minimizes

$$\frac{1}{M} \sum_{j=1}^M \left[\left(e^{-r(\tau_{j,k+1}^* - t_k)} f(S_{\tau_{j,k+1}}^j) - \sum_{l=1}^L \alpha_{k,l} P_l(S_{t_k}^j) \right)^2 \right]$$

- For all $j \in \{1, \dots, M\}$ we define

$$\tau_{j,k} = t_k \mathbf{1}_{A_{j,k}} + \tau_{j,k+1} \mathbf{1}_{A_{j,k}^c}, \text{ for } 0 \leq k \leq n-1$$

where

$$A_{j,k} = \left\{ f(S_{t_k}^j) \geq \sum_{l=1}^L \alpha_{k,l}^j P_l(S_{t_k}^j) \right\}$$

Price approximation

- For $k = 0$ the price of the Bermudan option is approximated by

$$\frac{1}{M} \sum_{j=1}^M e^{-r\tau_{j,0}} f(S_{\tau_{j,0}}^j)$$

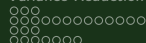
- The Longstaff Schwarz algorithm converges in $\mathcal{L}^2$ as $L \rightarrow +\infty$
- For a fixed L , it converges almost surely as $M \rightarrow +\infty$

Computing the coordinates $(\alpha_{k,l})_{1 \leq l \leq L}$ - Basic approach

- For a fixed time step t_k , this approach consists on simply solving the following system

$$\begin{pmatrix} P_1(S_{t_k}^1) & \dots & P_L(S_{t_k}^1) \\ \vdots & \vdots & \vdots \\ P_1(S_{t_k}^M) & \dots & P_L(S_{t_k}^M) \end{pmatrix} \begin{pmatrix} \alpha_{k,1} \\ \vdots \\ \alpha_{k,L} \end{pmatrix} = \begin{pmatrix} e^{-r(\tau_{1,k+1}-t_k)} f(S_{\tau_{1,k+1}}^1) \\ \vdots \\ e^{-r(\tau_{M,k+1}-t_k)} f(S_{\tau_{M,k+1}}^M) \end{pmatrix}$$

- Advantage : easy to implement
- Drawback : not a high accuracy



Computing the coordinates $(\alpha_{k,l})_{1 \leq l \leq L}$ - Optimal approach

- For a fixed time step t_k we aim at computing the sequence $(\alpha_{k,l}^j)_{1 \leq l \leq L}$ that minimizes

$$\frac{1}{M} \sum_{j=1}^M \left[\left(e^{-r(\tau_{j,k+1}^* - t_k)} f(S_{\tau_{j,k+1}}^j) - \sum_{l \geq 1}^L \alpha_{k,l} P_l(S_{t_k}^j) \right)^2 \right]$$

- We differentiate the above quantity with respect to α_{k,l_0} and solve

$$\sum_{j=1}^M \left(e^{-r(\tau_{j,k+1}^* - t_k)} f(S_{\tau_{j,k+1}}^j) - \sum_{l \geq 1}^L \alpha_{k,l} P_l(S_{t_k}^j) \right) P_{l_0}(S_{t_k}^j) = 0$$

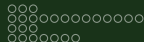
Computing the coordinates $(\alpha_{k,l})_{1 \leq l \leq L}$ - Optimal approach

- This is equivalent to solving

$$\sum_{l=1}^L \left(\sum_{j=1}^M P_l(S_{t_k}^j) P_{l_0}(S_{t_k}^j) \right) \alpha_{k,l} = \sum_{j=1}^M e^{-r(\tau_{j,k+1}-t_k)} f(S_{\tau_{j,k+1}}^j) P_{l_0}(S_{t_k}^j)$$

- Let $H_{l_0,l} = \sum_{j=1}^M P_l(S_{t_k}^j) P_{l_0}(S_{t_k}^j)$. We need to solve

$$\sum_{l=1}^L H_{l_0,l} \alpha_{k,l} = \sum_{j=1}^M e^{-r(\tau_{j,k+1}-t_k)} f(S_{\tau_{j,k+1}}^j) P_{l_0}(S_{t_k}^j)$$



Computing the coordinates $(\alpha_{k,l})_{1 \leq l \leq L}$ - Optimal approach

We can write this in a matrix equation given by

$$H_{\alpha_k} = \sum_{j=1}^M e^{-r(\tau_{j,k+1} - t_k)} f(S_{\tau_{j,k+1}}^j) P(S_{t_k}^j)$$

where

$$\alpha_k = (\alpha_{k,1}, \dots, \alpha_{k,L}) \quad \text{and} \quad P(S_{t_k}^j) = (P_1(S_{t_k}^j), \dots, P_L(S_{t_k}^j))$$